

2024 SURVEY

Developer Workforce & Compensation Trends

Stack Overflow Developer Survey

65,437

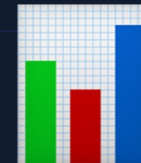
Respondents

180+

Countries

2024

Edition



STACK OVERFLOW

EMPLOYMENT STATUS

84% of Developers Are Working, but Full-Time Employment Dips in the U.S.

KEY INSIGHT: U.S. full-time employment fell 4pp year-over-year to 64.8% — plan for greater contractor and freelancer reliance
22,677 salary responses | 34.7% response rate | 65,437 total respondents

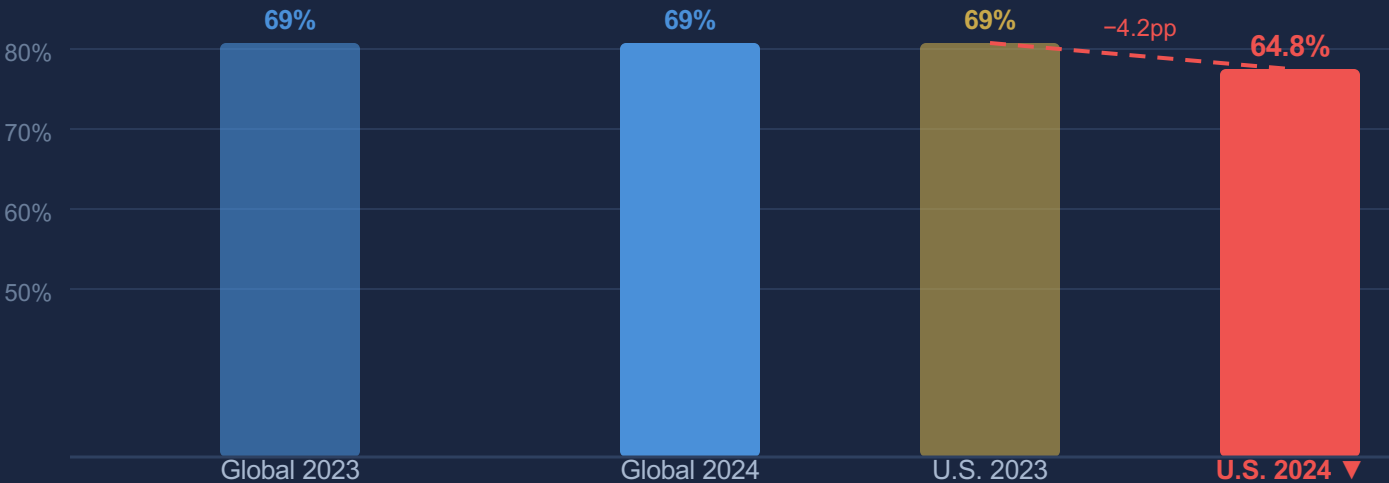
Full-Time (Global)
69% ▼ 4.2pp U.S.

Freelance / Contract
16.4%

Full-Time Students
13.2%

Currently Working (any mode)
84% of 65,437 respondents

Full-Time Employment: Global vs. U.S. (2023 → 2024)



Regional Employment Patterns

United States
Full-time 64.8% (↓ from 69% in 2023)

U.K. & Ukraine
Non-full-time → lean toward contracting

Germany & India
Non-full-time → lean toward student status

Strategic Implication
As full-time employment softens in the U.S., HR teams should build scalable contractor pipelines and adjust headcount planning models accordingly.

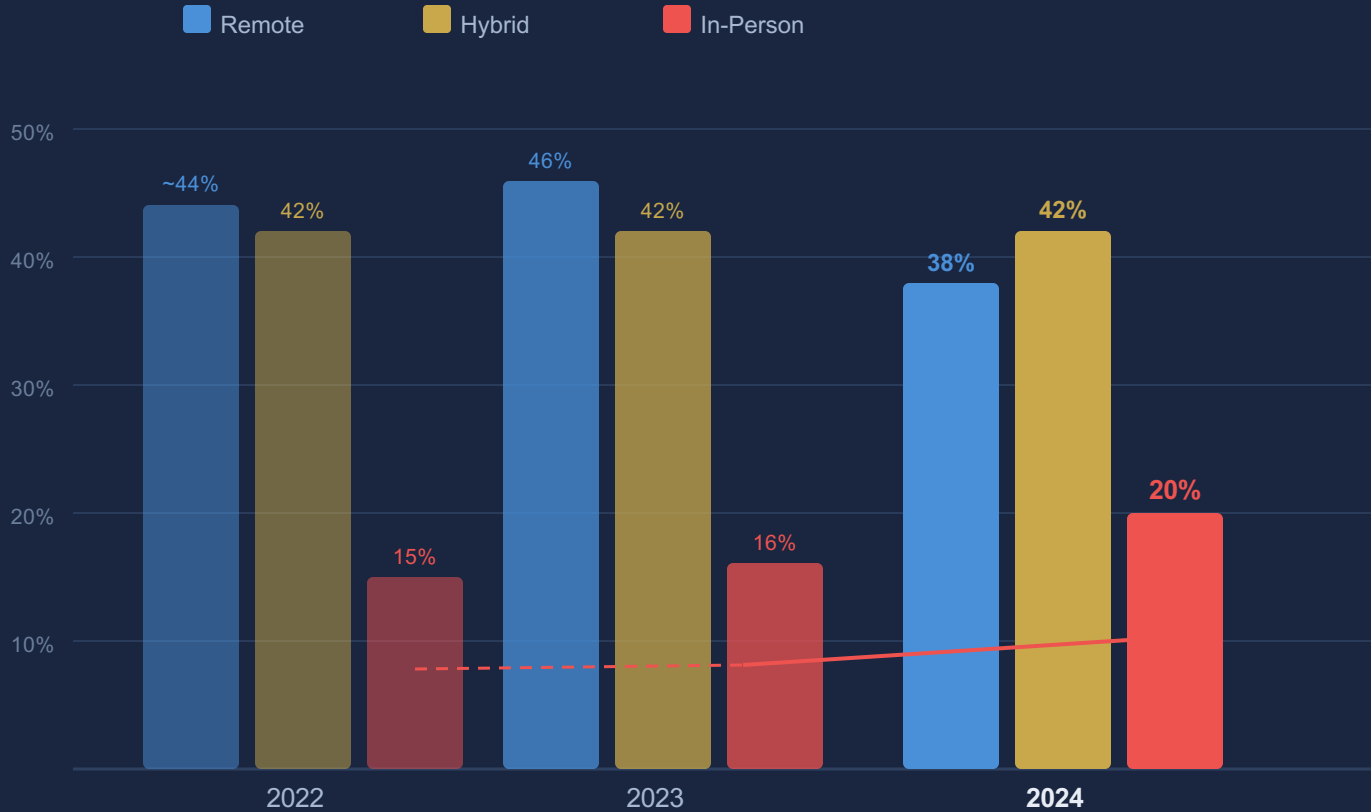
WORK ENVIRONMENT

In-Person Work Rises to 20% While Hybrid Holds Steady at 42%

KEY INSIGHT: Third consecutive year of in-person growth (15%→16%→20%) — RTO trend is real, though hybrid (42%) still dominates

Hybrid work is most common at 42% | Remote at 38% | In-person at 20% in 2024

Work Environment Trends — 2022 to 2024



2024 Snapshot

42%

Hybrid — most common arrangement stable vs. 2023

38%

Remote — second most common ↓ from ~46%

20%

In-Person — rising for 3rd year ↑ +5pp since 2022

What This Means for HR Leaders

Hybrid remains the expectation, not the exception. RTO mandates risk talent attrition — especially for top technical talent who have leverage in the current market.

Structure hybrid intentionally: define office days with purpose, not mandates.

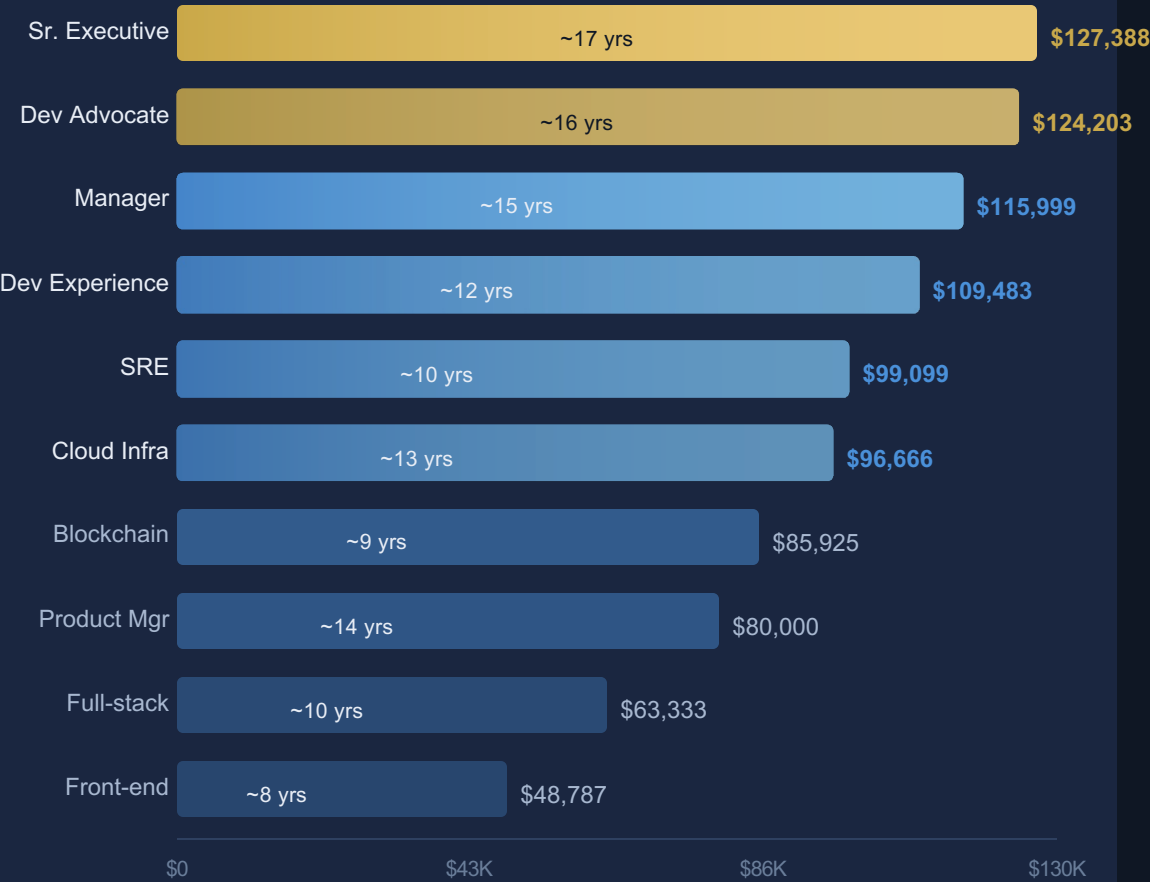
SALARY BY DEVELOPER TYPE

Senior Executives and Dev Advocates Top \$124K+ Median Salary

KEY INSIGHT: SRE (\$99K) and Cloud Infra (\$97K) are the highest-paid purely technical roles; experience premium is steep above 15 years

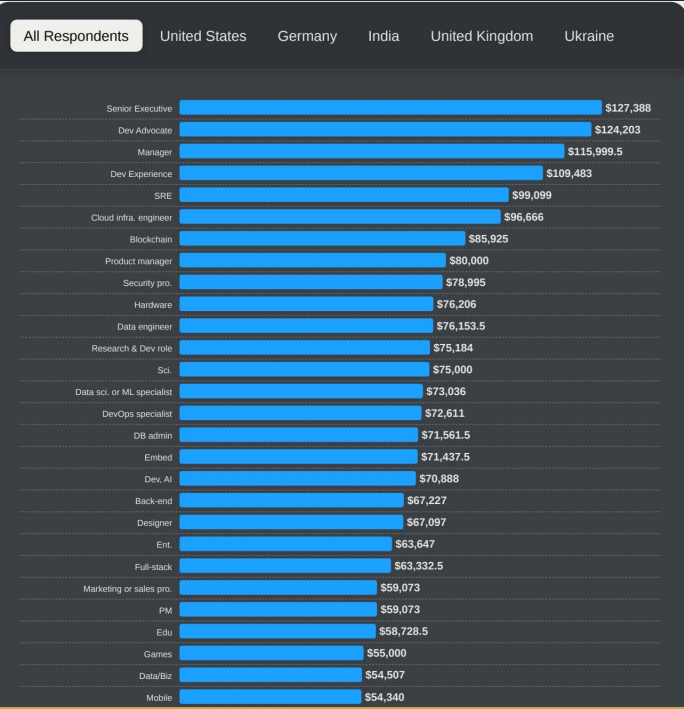
22,677 salary responses | 34.7% response rate | Median annual compensation (USD)

Median Salary by Developer Type (USD)



Bar label shows avg. years of experience | Top 10 of 11 developer types shown

Survey Chart Reference



Regional Note

Mobile devs earn more in the U.S.
Data engineers lead in Germany & Ukraine

Sample Size

22,677 salary responses
34.7% overall response rate

EXPERIENCE VS. SALARY

Experience Pays: Top Roles Combine 14+ Years with Six-Figure Salaries

KEY INSIGHT: Senior roles cluster in the upper-right (high exp + high pay); Student is a stark outlier at \$15K / 2 years experience

Scatter plot: x = avg. years of experience | y = median salary (USD) | Each point = one developer type

Role Clusters by Experience & Salary

PREMIUM CLUSTER — 15+ yrs exp | \$116K–\$127K

Sr. Executive	\$127,388	~17 yrs
Dev Advocate	\$124,203	~16 yrs
Manager	\$115,999	~15 yrs

Pattern: Leadership & advocacy roles. Experience premium is highest in this tier.

TECHNICAL SENIOR — 10–13 yrs exp | \$97K–\$109K

Dev Experience	\$109,483	~12 yrs
SRE	\$99,099	~10 yrs
Cloud Infra	\$96,666	~13 yrs

Pattern: Reliability & infrastructure specialists — high leverage, high compensation.

MID-RANGE — 8–10 yrs exp | \$49K–\$86K

Full-stack \$63K | Front-end \$49K | Blockchain \$86K

Largest talent pool; significant variation within tier by specialization

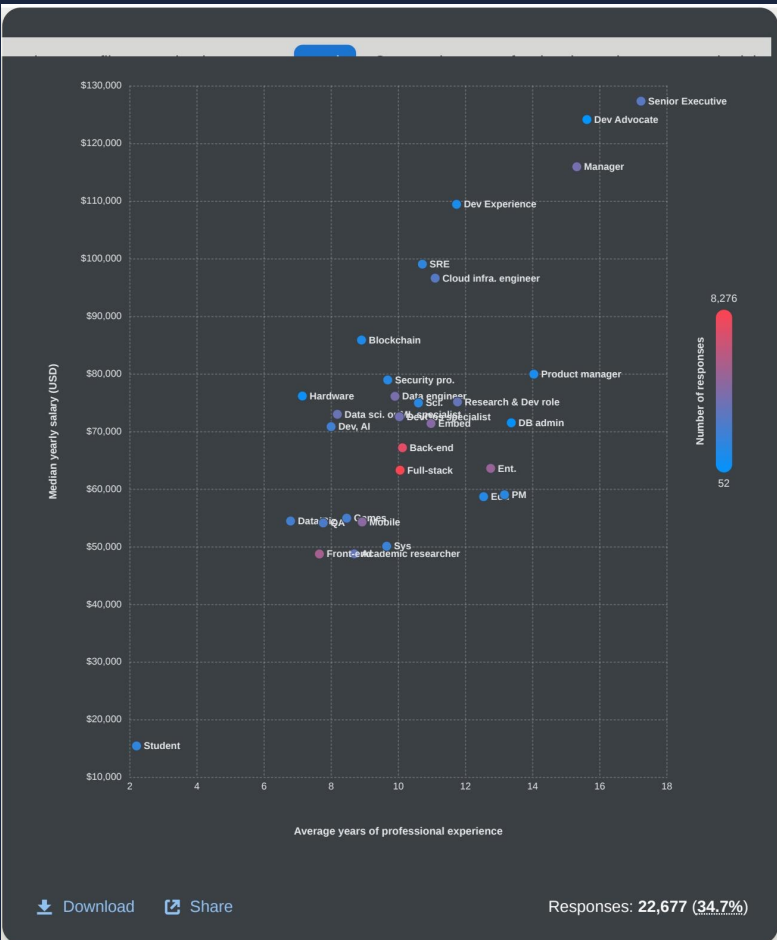
OUTLIER: Student — \$15,466 | ~2 yrs exp

Benchmark context: Student salary is 8x lower than the Sr. Executive median.
Represents entry-point; useful for internship and new-grad compensation anchoring.

Benchmark Takeaway for Compensation Bands

Experience is the strongest predictor of comp. Set bands at 10yr, 15yr milestones.
Flat bands across experience levels will drive attrition of senior talent.

Survey Scatter Plot Reference (figure_2)



Y-axis: Median salary (USD) | X-axis: Avg. years of experience | Each point = developer type

SALARY BY PROGRAMMING LANGUAGE

Erlang & Clojure Developers Earn >\$95K; Dart & Prolog Below \$45K

KEY INSIGHT: Niche functional languages (Erlang, Clojure, Elixir) command \$30K+ premium over mainstream JS/PHP — factor language into comp bands

Overall salary range dropped ~\$10K vs. 2023 (now \$60–75K avg vs. \$70–85K last year)

Language Salary Tiers (Median USD)

TIER 1 — NICHE FUNCTIONAL | \$93K–\$97K

Erlang	~\$97,000	~12 yrs
Clojure	~\$95,000	~11 yrs
Elixir	~\$93,000	~11 yrs

Niche adoption = scarcity premium. High experience requirement reinforces premium.

TIER 2 — ESTABLISHED NICHE | \$88K–\$90K

Perl ~\$90K (~19 yrs)	Ruby ~\$89K (~11 yrs)
Scala ~\$88K (~9 yrs)	

Perl's high salary reflects extreme experience depth (19yr avg).

TIER 3 — MAINSTREAM | \$64K–\$76K

Go ~\$76K (~9 yrs)	Python ~\$67K (~9 yrs)
JavaScript ~\$64K (~10 yrs)	

High supply depresses wages; Go commands premium for systems work.

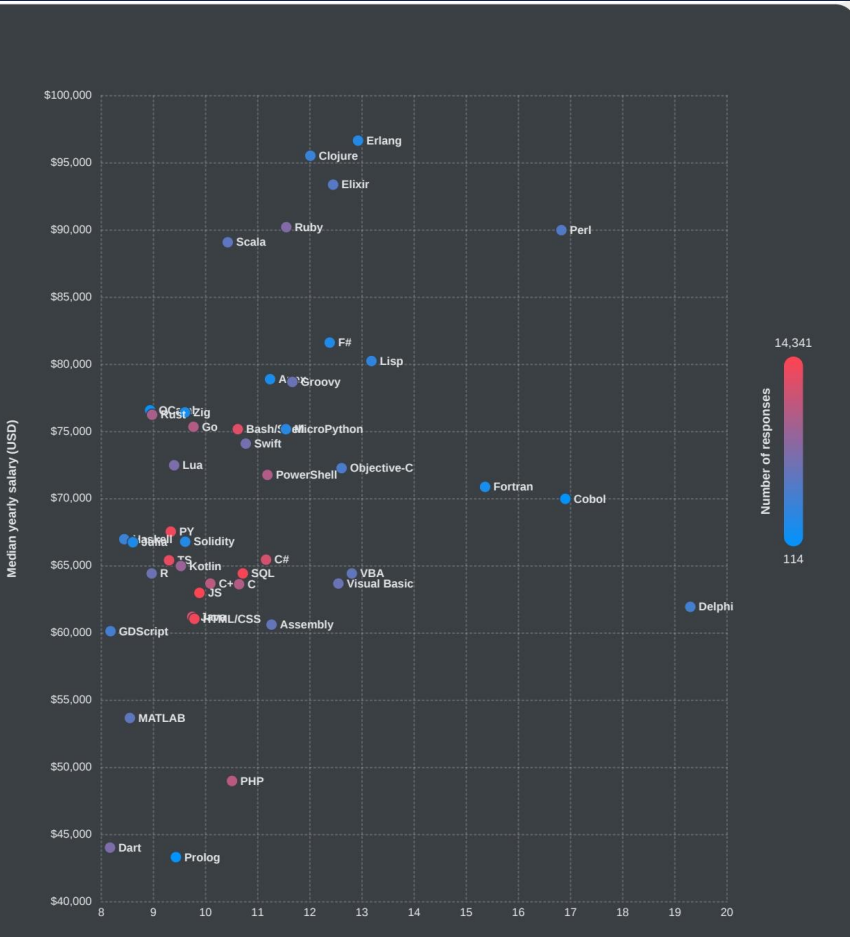
TIER 4 — LOWER EARNERS | \$43K–\$49K

PHP ~\$49K (~10 yrs)	Dart ~\$44K (~8 yrs)
Prolog ~\$43K (~9 yrs)	

PHP: large pool in low-cost markets. Dart: young ecosystem. Prolog: niche academic.

▼ Overall salary range dropped ~\$10K vs. 2023 now \$60–75K avg vs. \$70–85K in 2023

Survey Scatter Plot Reference (figure_3)



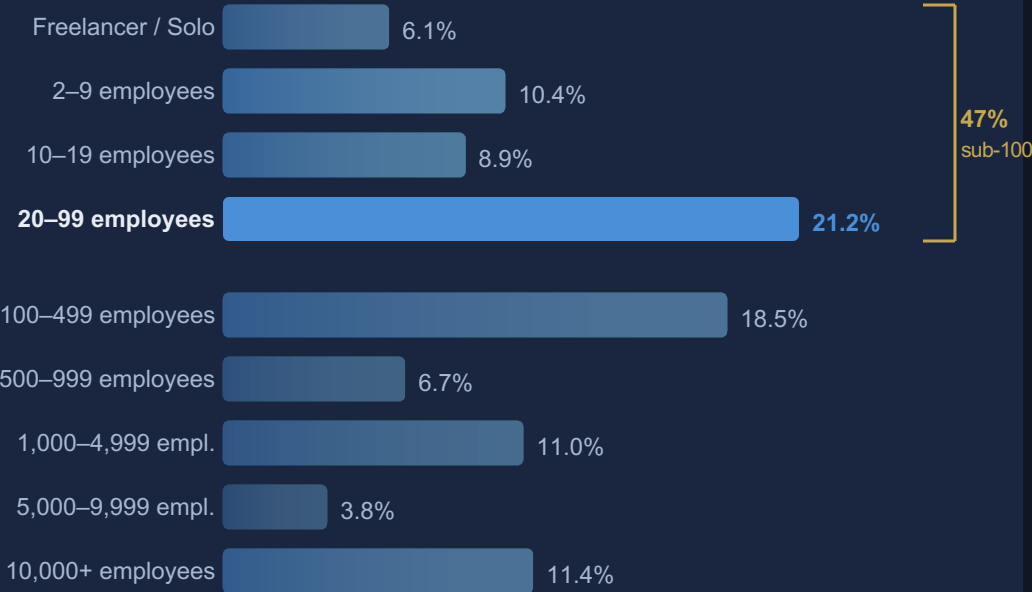
Y-axis: Median salary (USD) | X-axis: Avg. years of experience | Each point = programming language

COMPANY SIZE & PURCHASING INFLUENCE

47% Work at Sub-100 Orgs; 62% Influence Tech Purchases

KEY INSIGHT: Individual contributors — not just management — drive vendor decisions: engage developers directly in your tooling evaluations
42,827 respondents for company size | 62% of all respondents influence tech purchasing

Company Size Distribution (42,827 respondents)



Context for Hiring Strategy

Nearly half of developers work in small orgs — compensation at these firms may lag

Enterprise (1,000+): 26.2% of market — larger budgets but slower hiring cycles
SMB (2–99): 40.5% — often offer equity / flexibility to compete on total comp

Technology Purchasing Influence

62% of developers influence tech purchasing decisions
Up and down the org chart

Influence Rate by Role



Preferred Buying Approach

60% prefer Build-and-Buy (hybrid approach)
vs. pure build-in-house or pure buy (off-the-shelf)

Top Evaluation Methods

- 1. Start a free trial 75% of respondents
 - 2. Ask other developers 73% of respondents
- Vendor implication: developer-led trials and peer reviews are the primary purchase drivers

KEY TAKEAWAYS

What These Trends Mean for Hiring & Retention

BOTTOM LINE: Act on these five signals to stay ahead in developer talent acquisition and retention in 2024–2025

Based on 65,437 respondents across 180+ countries | Stack Overflow Developer Survey 2024

01 — Full-Time Employment Softening

U.S. full-time employment fell from 69% to 64.8% in one year.
Build scalable contractor pipelines alongside FTE hiring plans.
Freelance/contract remained strong at 16.4% globally — talent pool is growing.

→ Action: Revisit workforce mix ratios and contractor onboarding processes

02 — Hybrid Is the Expectation, Not Exception

42% hybrid (stable), in-person rising to 20% (3rd consecutive year up).
Blanket RTO mandates risk attrition of top technical talent.
Remote dropped from ~46% to 38% — some convergence on hybrid.

→ Action: Define structured hybrid — purpose-driven office days, not mandate-driven

03 — DevRel and SRE Roles Command Premium Pay

Sr. Executive (\$127K), Dev Advocate (\$124K), Manager (\$116K) top the chart.
SRE (\$99K) and Cloud Infra (\$97K) are the highest purely technical roles.
All premium roles share 15+ years average experience — experience premium is steep.

→ Action: Calibrate comp bands using experience milestones at 10yr and 15yr

04 — Language Choice Correlates with Pay

Erlang/Clojure/Elixir devs earn \$93K–\$97K vs. \$64K for JavaScript.
Overall salary range dropped ~\$10K year-over-year across all languages.
Niche language scarcity creates pay premiums — not just experience depth.

→ Action: Factor primary tech stack into offer benchmarks and bands

05 — Engage Developers in Vendor Decisions — They Drive the Choice

62% of all respondents influence tech purchasing — not just management. Engineering managers at 87%, Product managers at 77%.
60% prefer a build-and-buy approach. Top evaluation methods: free trial (75%) and asking other developers (73%).
This has dual implications: HR tools must win developer buy-in, and vendor outreach must target ICs not just leadership.

→ Action: Include senior ICs in tooling evaluations; prioritize free-trial-first procurement strategies

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